Router Configuration Commands

Static Routing • RIP v2 • OSPF

Configuration commands for the two-router GNS3 topology using FastEthernet interfaces, two switches, and two VPCS PCs.

1. Static Routing Configuration

R1 — Interface Configuration

R1 Interfaces

```
enable
configure terminal
interface fastethernet0/0
ip address 10.0.0.1 255.255.255.0
no shutdown
exit
interface serial0/0
ip address 10.0.1.1 255.255.255.0
no shutdown
exit
ip route 10.0.2.0 255.255.255.0 10.0.1.2
endip route 10.0.2.0 255.255.255.0 10.0.1.2
```

R2 Interfaces

```
enable
configure terminal
interface fastethernet0/0
ip address 10.0.2.1 255.255.255.0
no shutdown
exit
interface serial0/0
ip address 10.0.1.2 255.255.255.0
no shutdown
exit
ip route 10.0.0.0 255.255.255.0 10.0.1.1
end
```

R2 Static Route

```
ip route 10.0.0.0 255.255.255.0 10.0.1.1
```

PC1 IP Configuration

```
ip 10.0.0.2 255.255.255.0 10.0.0.1
```

PC2 IP Configuration

```
ip 10.0.2.2 255.255.255.0 10.0.2.1
```

Verification Commands

show ip interface brief

show ip route

Test Connectivity (PC1)

ping 10.0.2.2

2. RIP Version 2 Configuration

R1 — RIP v2

```
enable

configure terminal

interface fastethernet0/0

ip address 10.0.1.1 255.255.255.0

no shutdown

exit

interface fastethernet0/1

ip address 10.0.0.1 255.255.255.0

no shutdown

end

enable
configure terminal
router rip
version 2
no auto-summary
network 10.0.0.0
end
```

R2 — RIP v2

```
enable

configure terminal

interface fastethernet0/0
```

```
ip address 10.0.1.2 255.255.255.0

no shutdown

exit

interface fastethernet0/1

ip address 10.0.2.1 255.255.255.0

no shutdown

end

enable
configure terminal
router rip
version 2
no auto-summary
network 10.0.0.0
end
```

RIP Verification Commands (run on routers)

```
show ip interface brief
show ip route
show ip protocols
```

PC Connectivity Tests

```
PC1: ping 10.0.2.2
PC2: ping 10.0.0.2
```

3. OSPF Configuration

R1 — OSPF Process 1, Area 0

```
enable
configure terminal
router ospf 1
network 10.0.0.0 0.0.0.255 area 0
network 10.0.1.0 0.0.0.255 area 0
end
```

R2 — OSPF Process 1, Area 0

```
enable
configure terminal
router ospf 1
network 10.0.1.0 0.0.0.255 area 0
network 10.0.2.0 0.0.0.255 area 0
end
```

OSPF Verification Commands (run on routers)

```
show ip interface brief
show ip route
show ip ospf neighbor
```

PC Connectivity Tests

```
PC1: ping 10.0.2.2
PC2: ping 10.0.0.2
```

Addressing Reference

Device / Interface	IP Address	Subnet Mask / Gateway
PC1	10.0.0.2	255.255.255.0 / GW 10.0.0.1
R1 FastEthernet0/1	10.0.0.1	255.255.255.0
R1 FastEthernet0/0	10.0.1.1	255.255.255.0
R2 FastEthernet0/0	10.0.1.2	255.255.255.0
R2 FastEthernet0/1	10.0.2.1	255.255.255.0
PC2	10.0.2.2	255.255.255.0 / GW 10.0.2.1

Note: Configure and test one routing protocol at a time. If changing from RIP to OSPF or static routing, remove the previous routing configuration as appropriate to avoid confusion.